

DIAL ALERT Capstone Final Report

Early prediction of blood pressure defined intradialytic hypotension

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Analytic cohort	Primary event rate	Test ROC AUC	Test average precision
106,758 sessions	8.49%	0.852	0.395

Academic clinical decision support prototype. The model is not a diagnostic device and is not ready for clinical deployment.

This report consolidates problem framing, data understanding, preprocessing, exploratory analysis, feature engineering, model comparison, explainability, fairness auditing, business translation, and reproducibility.

Report Structure

The report follows the machine-learning lifecycle and the seven graded capstone steps. Each section distinguishes observed evidence from interpretation and proposed future work.

Section	Purpose
Executive Summary	Main result, operational meaning, and recommendation
1 Problem Understanding and Framing	Clinical context, task type, metrics, and business KPIs
2 Data Collection and Understanding	Provenance, cohort, variables, quality, and license
3 Preprocessing EDA and Feature Engineering	Cleaning, leakage control, relationships, selection, and PCA
4 Model Implementation and Comparison	Algorithms, grouped validation, locked test performance, and capacity
5 Critical Thinking Ethical AI and Bias Auditing	Explanations, limitations, fairness metrics, and mitigations
6 Final Communication and Business Translation	Workflow, ROI framework, risks, and pilot design
7 Reproducibility and GitHub Repository	Environment, artifacts, tests, CI, and repository structure
8 Conclusions and Next Steps	Decision, validation gates, and research priorities

Executive Summary

DIAL-ALERT tests whether information available early in a haemodialysis session can identify sessions that later develop a systolic blood-pressure reading below 90 mmHg. The project uses HEMOBP Version 3, a public longitudinal dataset released under CC BY 4.0. Deterministic linkage and eligibility rules produced 106,758 sessions from 830 patients, with 9,067 primary events. [1,2]

Seven supervised modelling configurations were compared using patient-grouped cross-validation and patient-disjoint validation. Random forest was selected under a locked rule that prioritized grouped-CV average precision and then validation Brier score when models were within 0.01 of the best result. The untouched test set contained 21,354 sessions from 170 patients.

On the locked test set, the selected model achieved average precision 0.395, ROC AUC 0.852, and Brier score 0.062. At the validation-selected threshold of 0.143, sensitivity was 66.1%, specificity 86.1%, and precision 30.6%. At a 20% alert capacity, the model captured 69.1% of observed events with 29.4% precision and 3.46-fold lift over prevalence.

The model concentrates retrospective risk but does not prove that an alert prevents an event or produces a financial return. The appropriate decision is to continue development through external validation and a governed prospective silent-mode pilot. Clinical use should remain prohibited until performance, calibration, workload, subgroup outcomes, and human factors meet prespecified acceptance criteria.

Decision area	Conclusion
Technical feasibility	Supported for continued validation; discrimination and risk concentration are promising
Clinical effectiveness	Unknown; no alert-triggered intervention was tested
Operational feasibility	Requires capacity testing because false alerts remain substantial
Fairness	Partially auditable; recorded-sex and age disparities require monitoring, while key attributes are absent
Deployment readiness	Not ready; external and prospective validation remain mandatory

1 Problem Understanding and Framing

Clinical and Operational Problem

Intradialytic hypotension can interrupt treatment and require rapid reassessment. Routine haemodialysis already generates demographic, session, machine, and blood-pressure data. DIAL-ALERT asks whether those existing data can support early risk ranking so that limited clinical attention is directed toward sessions with the greatest observed risk.

The intended user is a dialysis clinician or nurse. The proposed output is a probability and a capacity-aware alert, not a treatment order. The model should supplement observation and judgment; it should never suppress routine monitoring or direct an intervention without a clinician.

Data Science Task

Element	Prespecified definition
Task	Supervised binary classification
Analysis unit	One eligible patient-day haemodialysis session
Prediction time	Earliest valid active-dialysis observation during minutes 0 to 30
Target	Any later systolic blood-pressure measurement below 90 mmHg
Predictors	Index-time variables and patient history derived only from earlier sessions
Exclusions	Index SBP below 90 mmHg, inadequate later measurements, or last observed dialysis minute below 120

Success Measures

Type	Measure	Reason
Primary technical	Average precision	Assesses event ranking under 8.49% prevalence and is more informative than accuracy alone
Secondary technical	ROC AUC, Brier score, and log loss	Assesses ranking and probability quality
Threshold performance	Sensitivity, specificity, precision, and F1	Quantifies misses and false alerts at the operating threshold
Operations and business	Capacity metrics, review time, intervention rate, outcomes, and net cost	Measures workload and future real-world value without assuming savings
Equity	Selection, error, calibration, and disparity measures	Checks whether overall performance hides subgroup differences

2 Data Collection and Understanding

Provenance and License

HEMOBP Version 3 was selected because it links patient characteristics, session summaries, and time-stamped dialysis-machine measurements. The public release is versioned on Figshare, has stable checksums, and is licensed under CC BY 4.0. The peer-reviewed data descriptor documents the collection context. [1,2]

Source file	Grain	Rows	Verified MD5
idp.csv	Patient	1,072	31d269f59bf4acc719f392ad9164d08a
d1.csv	Dialysis session	165,986	7db3b48bae2c73e3ecd731dbcd452233
vip.csv	Time-stamped monitor record	4,366,298	ff1589610cf7aa3b3e01ba979b6bdc44

HEMOBP source tables and DIAL ALERT analytic unit

Repeated monitor records are reduced to one leakage-controlled row per eligible dialysis session.

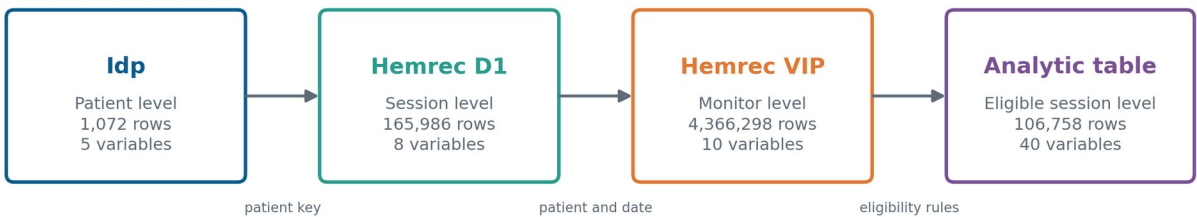


Figure 1. HEMOBP source tables and the DIAL-ALERT session-level analytic table.

Cohort Construction

The source tables are joined on patient identifier and normalized session date. The earliest valid active-dialysis reading in the first 30 minutes defines the index record. Later readings define outcomes. Sessions require at least two later measurement minutes and an observation at or beyond dialysis minute 120 (not 120 minutes after prediction).

Cohort stage	Count
Valid unique patient days	165,849
Patient days linked to monitoring	110,790
Sessions with an index record	108,781
Final eligible sessions	106,758
Unique analytic patients	830
Primary events	9,067

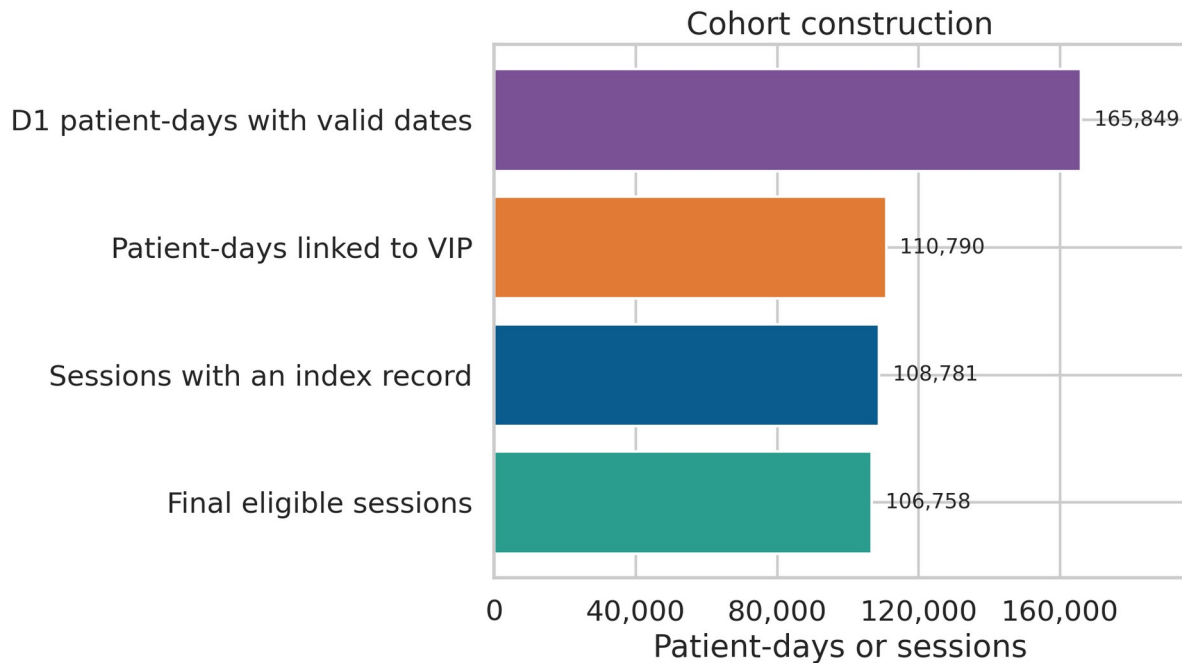


Figure 2. Deterministic cohort flow from valid session dates to the final eligible cohort.

Data Quality and Dictionary

Dates, linkage, duplicate patient-days, exact monitor duplicates, documented physiologic bounds, and follow-up completeness were audited before modelling. Missing values occur mainly in dialysis vintage and first-session history features. Plausible extremes were retained; removing unusual but possible dialysis sessions could hide clinically important risk.

The complete machine-readable dictionary in docs/data_dictionary.csv and docs/data_dictionary.md records each variable's source, data type, unit, allowed values, missingness, modelling role, and prediction-time availability. A separate feature-engineering dictionary documents the derived variables.

- The publication reports 1,075 outpatients, while the released patient file contains 1,072 unique rows; the analysis uses the actual public files and records this discrepancy.
- Symptoms, interventions, medication timing, laboratory data, and broad comorbidity data are unavailable.
- The target is therefore described as BP-defined IDH rather than a complete symptomatic clinical diagnosis.

3 Preprocessing EDA and Feature Engineering

Cleaning and Leakage Control

Step	Implementation and justification
Missing values	Median imputation for numeric variables and mode imputation for categorical variables inside each training fold
Duplicates	Stable removal of duplicate patient-days and exact duplicate monitor records
Outliers	Documented monitoring bounds remove impossible values; plausible extremes remain for audit and sensitivity analysis
Encoding	One-hot encoding for recorded sex and diabetes within the fitted pipeline
Scaling	Standardization for linear, L1, and PCA candidates; tree models retain original numeric scale
Temporal boundary	Only index-time variables and shifted prior-session history enter the feature matrix
Split boundary	Imputation, encoding, scaling, selection, PCA, and fitting occur within grouped training folds

Domain Derived Features

Derived variables translate raw dialysis records into quantities that can be interpreted clinically: fluid excess in kilograms and percent of dry weight, ultrafiltration intensity in mL/kg/h, mean arterial pressure, pulse pressure, dialysis vintage, and longitudinal history of prior IDH and nadir SBP. The count of prior sessions is log transformed for modelling.

Observed outcome rate across predictor deciles

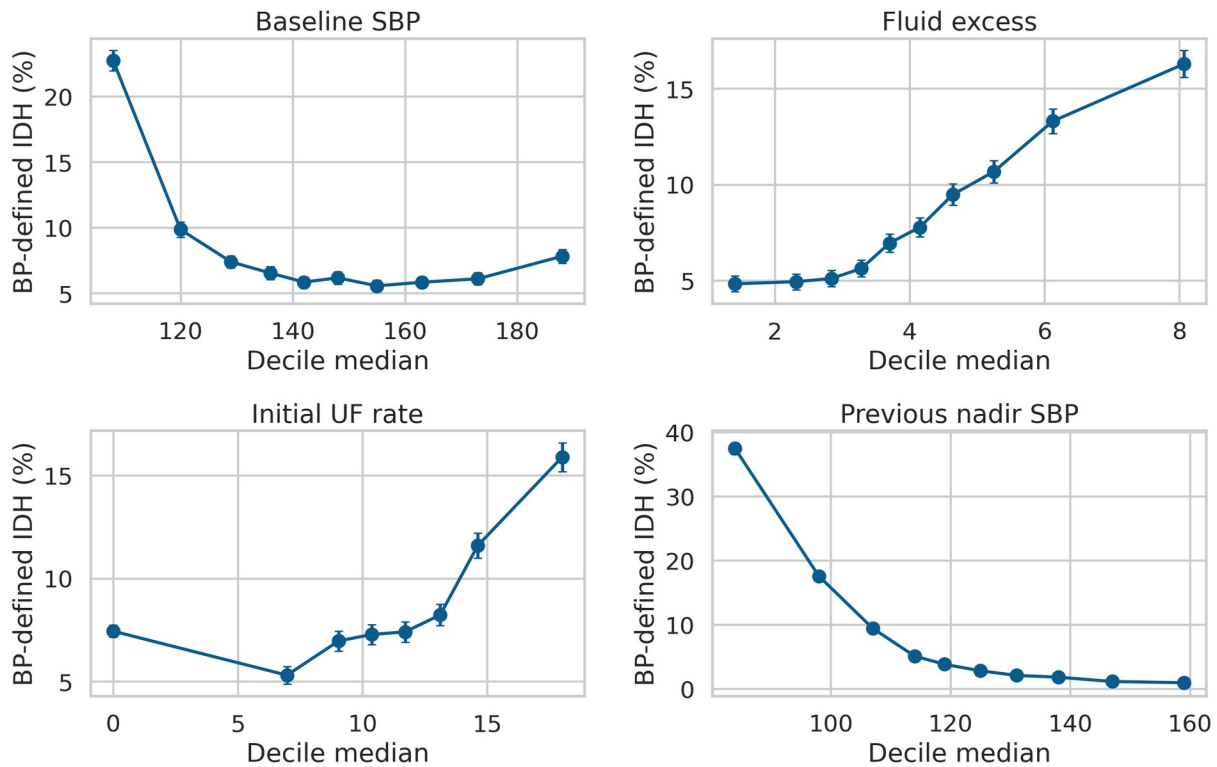


Figure 3. Observed outcome prevalence across bins of selected predictors. These are associations, not causal effects.

Feature Selection and Dimensionality Reduction

Embedded L1 logistic regression retained 12 of 24 transformed predictors. This candidate tested whether sparse selection could simplify the model without using test outcomes. PCA was fitted after numeric imputation and standardization; 11 components explained 87.6% of numeric-feature variance, exceeding the prespecified 85% threshold.

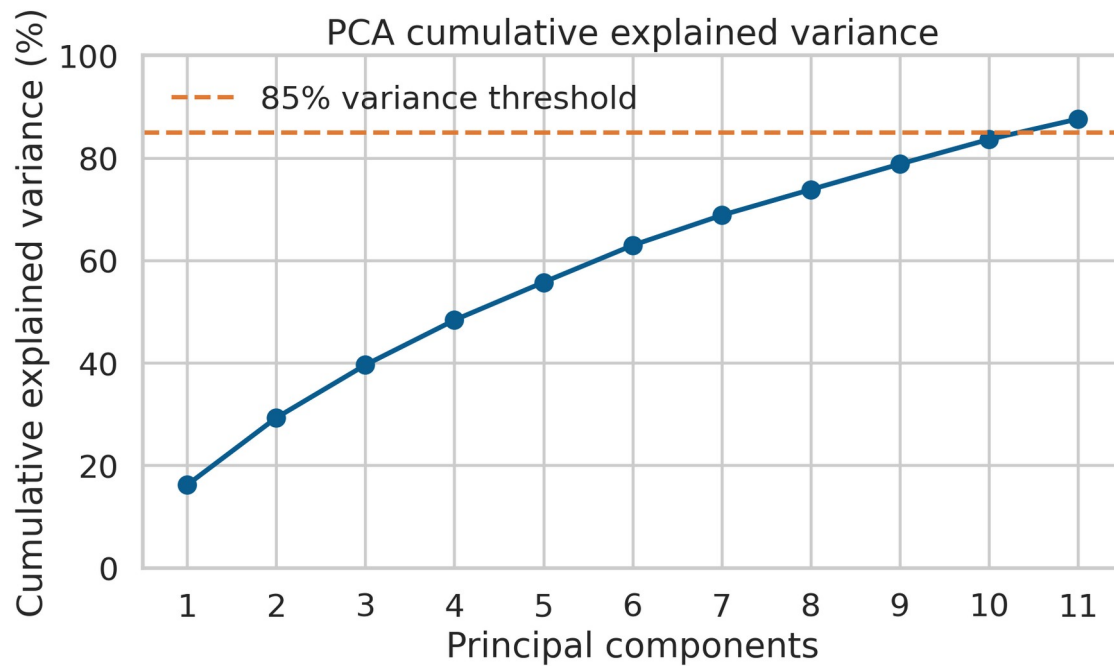


Figure 4. Cumulative explained variance of the numeric PCA candidate.

PCA was retained as a required dimensionality-reduction experiment rather than assumed to be beneficial. Its logistic-regression candidate had lower grouped-CV average precision than the selected random forest, showing that a lower-dimensional representation did not improve the primary objective.

4 Model Implementation and Comparison

Candidate Models and Validation

The comparison includes a prevalence baseline, logistic regression, embedded L1 feature selection, decision tree, PCA logistic regression, random forest, and histogram gradient boosting. Deep learning was not added because the inputs are structured tabular variables, there are only 830 independent patients, and calibration and interpretability are more important than model complexity.

DIAL ALERT inference workflow

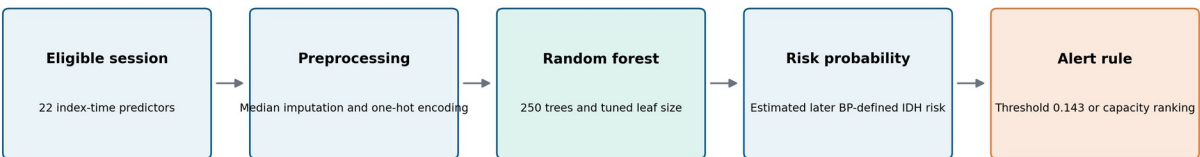


Figure 5. End-to-end model workflow from eligible session predictors to probability and alert output.

Partition	Patients	Sessions	Events	Prevalence
Training	496	64,053	5,440	8.49%
Validation	164	21,351	1,813	8.49%
Test	170	21,354	1,814	8.49%

Model Selection

Histogram gradient boosting had the highest grouped-CV average precision at 0.446, while random forest achieved 0.440. Because the two were within the prespecified 0.01 tolerance, validation Brier score determined the choice. Random forest had a validation Brier score of 0.058 compared with 0.134 for histogram gradient boosting and was selected.

Model selection balances ranking and calibration

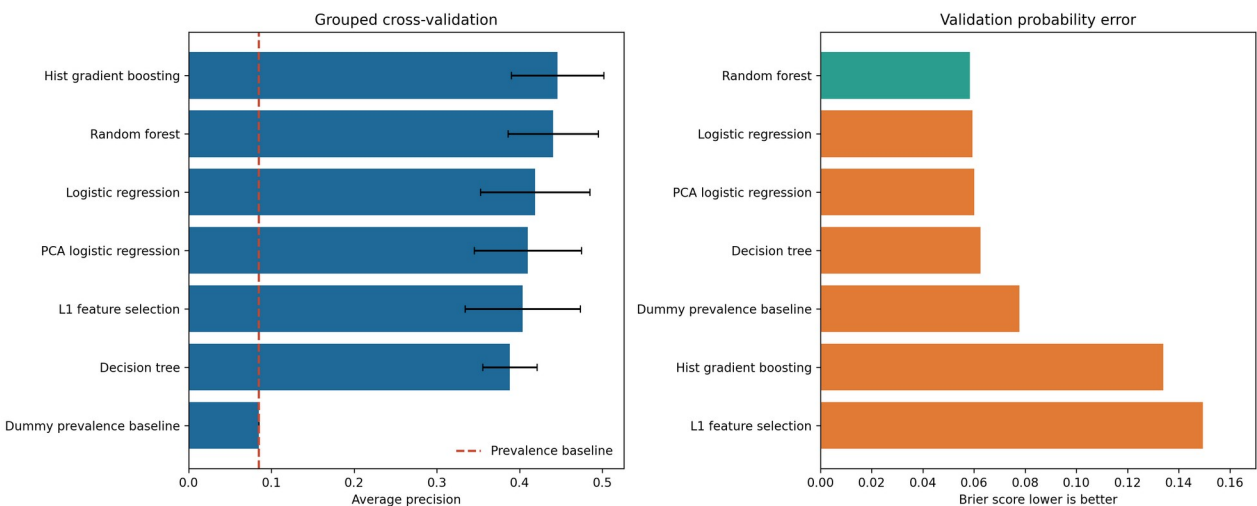


Figure 6. Candidate model comparison using grouped-CV average precision and validation Brier score.

Locked Test Performance

Metric	Estimate	Patient-bootstrap 95% interval
Average precision	0.395	0.306 to 0.478
ROC AUC	0.852	0.821 to 0.876
Brier score	0.062	0.049 to 0.077
Sensitivity	0.661	0.550 to 0.745
Specificity	0.861	0.806 to 0.903
Precision	0.306	0.260 to 0.353

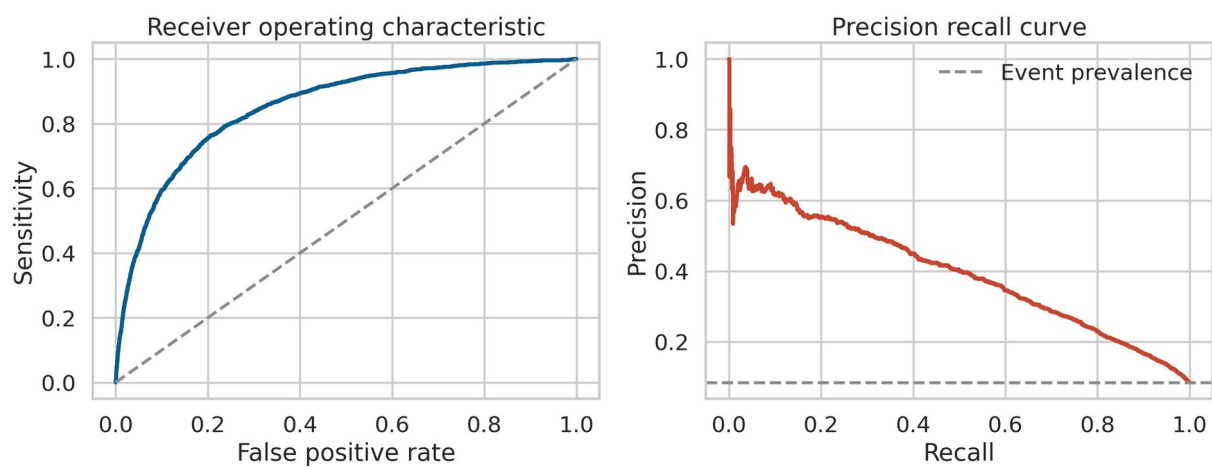


Figure 7. Locked test ROC and precision-recall curves.

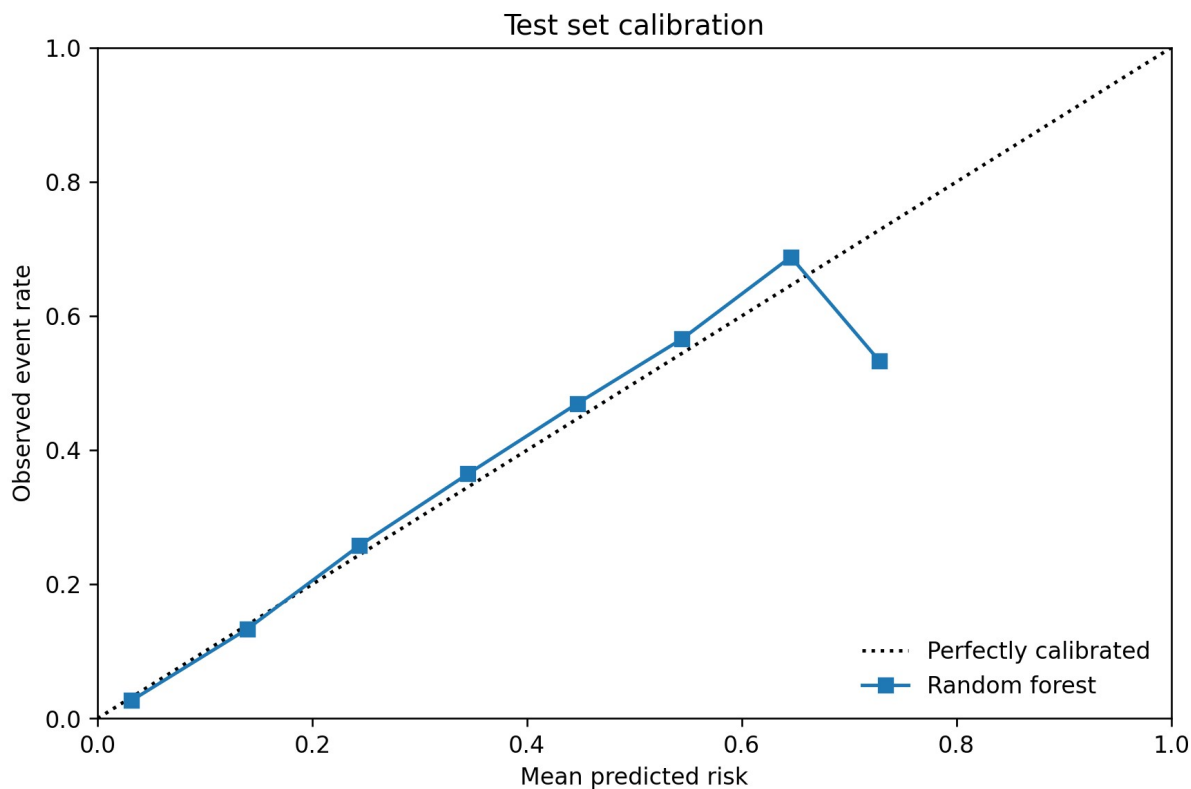


Figure 8. Test calibration curve. Sparse high-risk bins require cautious interpretation.

Operational Capacity

At the validation-selected probability threshold of 0.143, the model alerts on 18.3% of test sessions. A separate ranking analysis fixes the review capacity at 20% of sessions. Under that constraint, the highest-risk group contains 69.1% of events, but 14.1 non-event sessions per 100 total sessions would also be reviewed.

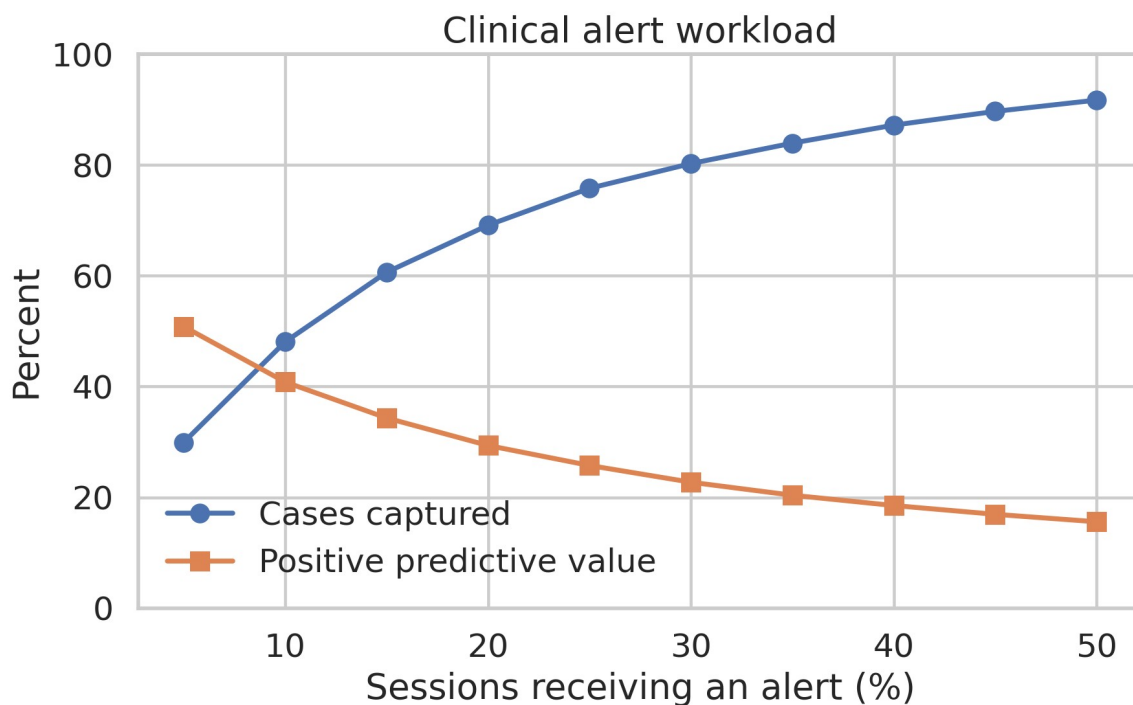


Figure 9. Recall, precision, lift, and false-alert workload across review capacity levels.

5 Critical Thinking Ethical AI and Bias Auditing

Model Explanations

Permutation importance, aggregate approximate interventional SHAP values, synthetic SHAP and LIME local explanations, and dependence curves across synthetic reference profiles provide complementary views of the fitted model without publishing patient-level explanation records. Prior-session nadir SBP and prior IDH rate were the strongest permutation features. Explanation outputs describe learned associations; they do not identify safe treatment targets or causal mechanisms. [5-7]

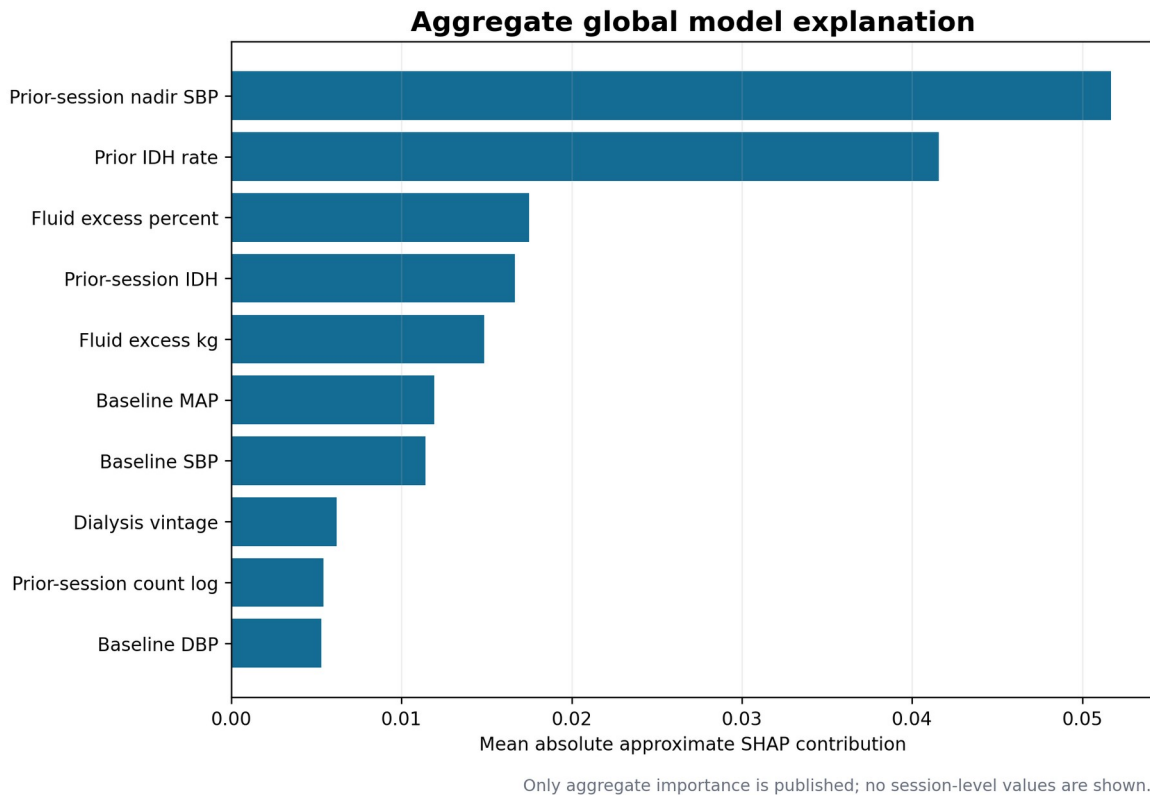


Figure 10. Aggregate approximate interventional SHAP importance for the locked model.

Synthetic-profile dependence and conditional effects

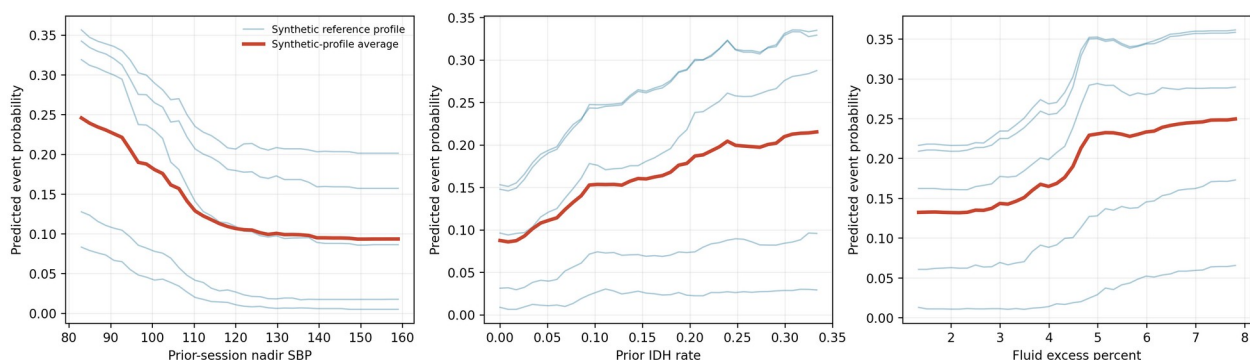


Figure 11. Dependence and conditional-effect patterns across synthetic reference profiles.

Limitations and Robustness

Risk	Evidence	Response
Class imbalance	Test prevalence is 8.49%	Use average precision, precision, sensitivity, calibration, and workload rather than accuracy alone
Patient leakage	No patient overlaps training, validation, and test	Retain grouped splitting in every future evaluation
Temporal leakage	Outcomes occur after the index time; history is shifted	Preserve the feature timestamp contract in production
Optimism	Grouped-CV AP 0.440, validation AP 0.470, test AP 0.395	Report patient-bootstrap uncertainty and require external validation
Repeated sessions	21,354 test sessions come from 170 patients	Use patient-cluster bootstrap and patient-equal-weight sensitivity analyses
Outcome validity	Symptoms and interventions are unavailable	Describe the endpoint as BP-defined IDH

Fairness Audit

Attribute	Disparate impact ratio	Sensitivity gap	FPR gap	Brier gap
Recorded sex	0.705	0.076	0.049	0.014
Age group	0.541	0.105	0.094	0.021
Diabetes status	0.988	0.110	0.010	0.009

Recorded sex and age are the only available protected-characteristic audits. Diabetes is a clinical subgroup and is not used as a socioeconomic proxy. Race, ethnicity, socioeconomic status, and gender identity are absent, so fairness across those groups cannot be assessed. The 0.80 disparate-impact ratio is treated as a descriptive screening heuristic, not a fairness certificate. [8]

Subgroup performance on the patient-disjoint test set

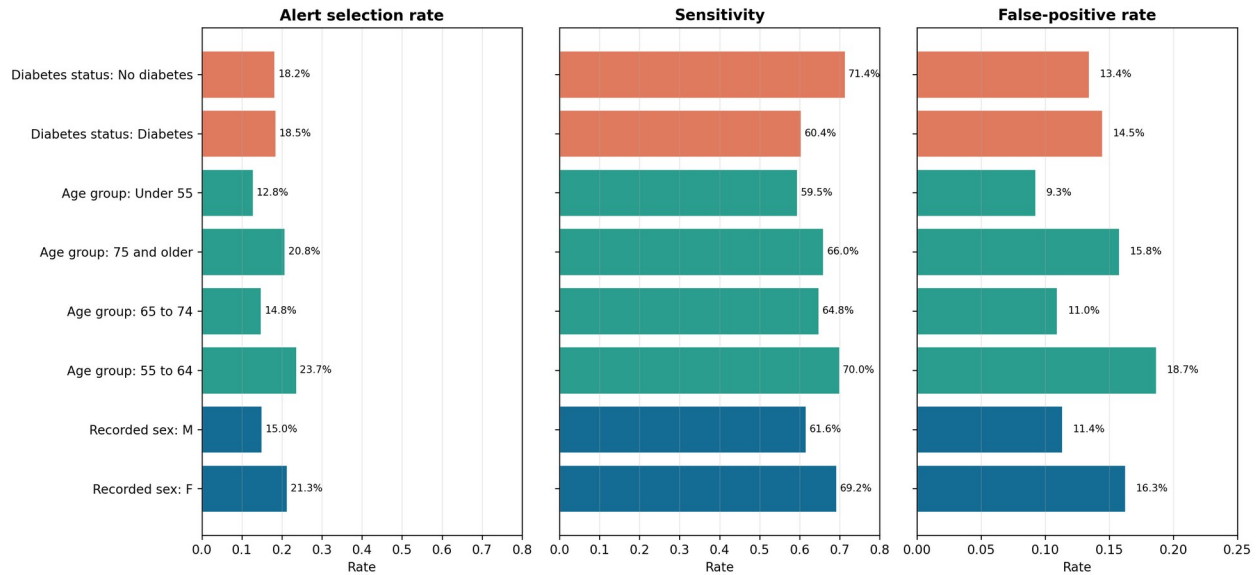


Figure 12. Selection and error metrics across available groups.

Mitigation Experiments

Strategy	Average precision	Precision	Sex impact ratio	Equalized odds gap
Original model	0.395	0.306	0.705	0.076
Sex outcome reweighting	0.394	0.309	0.764	0.058
Sex specific thresholds	0.395	0.278	0.665	0.139

Outcome-by-sex training weights improved the recorded-sex impact ratio from 0.705 to 0.764 and reduced the equalized-odds gap from 0.076 to 0.058, while average precision changed from 0.395 to 0.394. Sex-specific thresholds worsened held-out fairness and precision and are rejected. Reweighting remains a candidate for external testing, not an automatic deployment choice.

Mitigation changes fairness and performance together

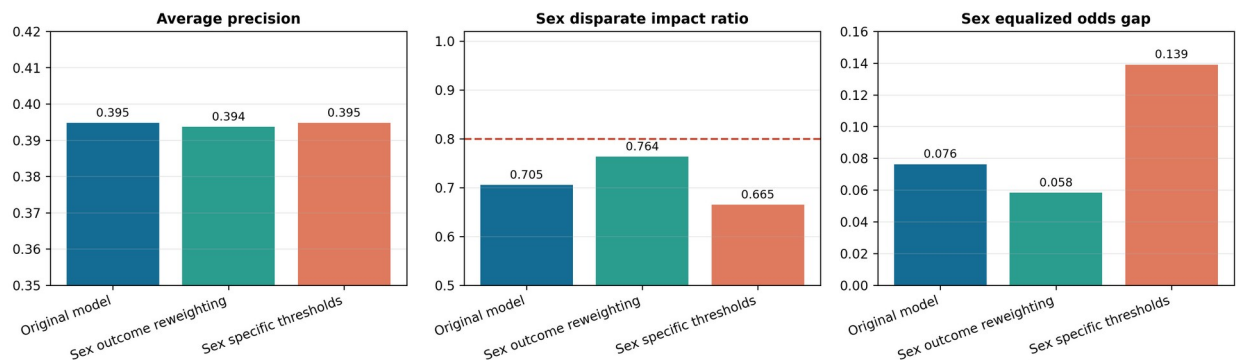


Figure 13. Performance and fairness tradeoffs across mitigation strategies.

6 Final Communication and Business Translation

Proposed Workflow

A future silent-mode pilot would calculate risk after the index observation, place the highest-risk sessions on a review list, and compare predictions with subsequent BP measurements without changing care. Only after prespecified safety and workload review should a later phase test whether presenting alerts changes clinician action or patient outcomes.

Per 100 eligible sessions at 20% capacity	Retrospective estimate
Sessions selected for review	20.0
Observed events in the cohort	8.5
Observed events concentrated in the selected group	About 5.9
Observed events outside the selected group	About 2.6
Selected sessions without the event	About 14.1

These values describe risk concentration in retrospective data. They are not prevented events. Every selected session still requires clinical interpretation, and some unselected sessions will experience the outcome.

ROI Measurement Framework

The pilot should measure net value rather than assume it. A transparent calculation is: net pilot value equals the number of events causally prevented multiplied by the local value of avoiding one event, less review time, implementation, maintenance, and monitoring costs. The prevention rate and local cost inputs are currently unknown.

Input	Pilot measurement
Alert workload	Alerts per 100 sessions and median review minutes
Clinical response	Proportion reviewed, action taken, action type, and override reason
Patient outcome	BP-defined events, symptoms, interventions, interrupted treatment, and recovery time
Safety	Missed events, delayed treatment, inappropriate action, and incident reports
Equity	Selection, sensitivity, false-positive rate, precision, and calibration by governed groups
Cost	Staff time, implementation, infrastructure, training, maintenance, and event-related utilization

Governance and Pilot Gates

Gate	Minimum evidence before progression
External validity	Acceptable discrimination and calibration in a different centre or later time period
Operational fit	Review volume and response time remain within agreed staffing capacity
Safety	No unacceptable harm signal, with defined clinician override and escalation
Fairness	Prespecified subgroup metrics and uncertainty reviewed by governance
Human factors	Clinicians understand the score, avoid automation bias, and document decisions
Monitoring	Data quality, drift, calibration, alert burden, incidents, and rollback are operational

The final communication package includes a sixteen-slide technical deck for peers and an eleven-slide business deck for executives. Both decks state the evidence boundary and recommend validation rather than deployment.

7 Reproducibility and GitHub Repository

The public repository is organized as an open-source project. Raw and processed data are excluded from Git because the source is large and patient-level. A version-pinned acquisition script downloads HEMOBP Version 3 and verifies MD5 checksums. The selected compressed predictor is retained so that inference can be tested without retraining.

Repository path	Contents
src	Acquisition, cohort construction, EDA, training, audit, inference, and report builders
notebooks	Guided end-to-end workflow
data	Acquisition instructions and empty tracked directories
models	Selected predictor, threshold specification, and manifest
artifacts	Metrics, uncertainty intervals, audit tables, and figures
docs	Data dictionary, preprocessing specification, and model card
reports	Final report, component reports, and technical and business presentations
tests	Contract and saved-model inference tests
.github workflows	Automated tests on each push and pull request

Reproduction Commands

```
python -m pip install -r requirements.txt
python src/download_data.py
python src/build_session_dataset.py --raw-dir data/raw --output-dir data/processed
python src/train_evaluate.py --data data/processed/hemobp_session_level.csv.gz --config configs/model_config.json --
artifacts artifacts --models models
python src/generate_eda.py
python src/generate_step4_assets.py
python src/audit_bias_fairness.py
python src/create_final_report.py
pytest -q
```

The model manifest records the data and configuration SHA-256 hashes, Python and package versions, selected model, random seeds, and hashes of saved artifacts. Continuous integration runs repository and prediction smoke tests. The notebook presents the same sequence with explanatory text.

8 Conclusions and Next Steps

DIAL-ALERT demonstrates the complete machine-learning lifecycle on a real, industry-relevant dataset. The project frames an auditable clinical question, constructs a leakage-controlled cohort, compares multiple models, evaluates an untouched patient-disjoint test set, explains model behavior, audits available groups, tests mitigation, and translates performance into operational workload.

The main finding is useful but limited: the random forest concentrates 69.1% of observed BP-defined events within the highest-risk 20% of sessions. This supports further validation. It does not demonstrate prevented events, safer dialysis, reduced cost, or transportability to another centre.

Recommended Sequence

- Validate the locked model on an external centre and a later time period without retuning.
- Collect symptoms, interventions, medication timing, broader comorbidities, and governed demographic attributes.
- Run a prospective silent-mode study to measure data availability, calibration, workload, and drift.
- Prespecify subgroup and safety thresholds, clinician override, incident response, and rollback rules.
- Only then test whether visible alerts change actions and improve patient outcomes in a controlled implementation study.

Final Decision

Continue DIAL-ALERT as a validation-stage clinical decision-support research project. Do not deploy it for patient-care decisions at this stage.

References

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Appendix Project Evidence Map

Project step	Evidence
1 Problem Understanding and Framing	Clear clinical context, supervised task, target, technical metrics, operational KPIs, and evidence boundary
2 Data Collection and Understanding	Peer-reviewed public dataset, license, checksums, cohort audit, complete dictionaries, and limitations
3 Preprocessing EDA and Feature Engineering	Nulls, duplicates, outliers, encoding, scaling, domain features, visual EDA, L1 selection, PCA, and explainability
4 Model Implementation and Comparison	Seven configurations, grouped tuning, locked selection rule, saved models, test metrics, intervals, and workload analysis
5 Critical Thinking Ethical AI and Bias Auditing	SHAP, local surrogate, PDP and ICE, leakage and overfitting analysis, fairness metrics, uncertainty, and mitigation tests
6 Final Presentation and Communication	Separate technical and business decks with notes, citations, ROI framework, risks, strategy, and pilot recommendation
7 GitHub Profile and Upload	Open-source structure, README, license, citation file, data acquisition, notebooks, tests, CI, final report, and reproducible commands

Submission interpretation and evidence

Outcome and eligibility

Predict any SBP below 90 mmHg after the earliest valid active-dialysis observation in minutes 0 to 30. Require index SBP at least 90, two distinct later measurement minutes, and last observed dialysis minute at least 120. The last requirement is anchored to dialysis time, not to prediction time. It is a retrospective eligibility criterion; short or interrupted sessions need separate prospective evaluation.

Distinct operating strategies

The fixed threshold is 0.142855878 (rounded to 0.143), chosen by maximum F2 on a patient-disjoint validation decision subset: test sensitivity 66.1%, precision 30.6%, 18.3 reviews and 12.7 false alerts per 100 sessions. Reviewing the highest-risk 20% instead gives 69.1% recall, 29.4% precision and 14.1 false alerts per 100. A prospective capacity policy must define the comparison batch and tie handling before use.

Model choice and uncertainty

Random forest was selected from candidates within 0.01 of the best grouped-CV average precision using validation Brier score. No post-hoc calibration was retained. An equally calibrated finalist comparison remains secondary work. The 21,354 test sessions come from only 170 patients. Patient-bootstrap 95% intervals: AP 0.306 to 0.478; ROC AUC 0.821 to 0.876; sensitivity at the fixed threshold 55.0% to 74.5%; recall at 20% capacity 63.2% to 74.4%.

Proposed future pilot targets

Provisional targets, not demonstrated prospective results: detect at least 65% of later SBP-below-90 events, generate no more than 15 false alerts per 100 eligible sessions, and achieve median review time at most 2 minutes per displayed alert. Prespecify one operating policy, denominators, uncertainty and safety stopping rules with the local team before evaluation.

Demonstration status and further research

Optional deployment evidence

Step 8 includes a local Flask app, a synthetic request, a recorded HTTP demo and instructions. The app and command-line route apply the same prior-session-count transformation. This is a local inference demonstration; no clinical or cloud deployment is established. Step 9 demonstrates saved AI-draft replay with numeric checks. Its published media does not demonstrate live LLM generation.

Reproducibility status

A new Python 3.12 environment was created on 18 September 2026. Source acquisition returned HTTP 403; end-to-end data-to-training reproduction therefore remains blocked. Included-model inference and application checks are recorded separately in docs/REPRODUCIBILITY_RECORD.md. Those checks do not verify the original training results.

Secondary analyses awaiting execution

Measure index-to-first-event warning time; compare a baseline-SBP plus prior-hypotension model; assess history ablation and first-observed sessions; compare both finalists with identical calibration splits and methods; and examine extreme UF/fluid-excess values and short-session exclusion. Prespecify analyses on development data, report them as exploratory, and do not replace the locked model based on repeated test-set comparisons.

Future validation sequence

Validate the locked model at another center or in a later period, confirm exploratory fairness mitigation on fresh patients, then run silent predictions. Only after data quality and safety gates pass should supervised clinician review assess workload, actions, unintended effects, outcomes and costs. A 90-day schedule is illustrative and cannot establish clinical or financial benefit by itself.